Mann Talati

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Education

University of Illinois Urbana-Champaign — **Expected Graduation: May 2028**
Master of Computer Science — *Champaign, IL*

University of Illinois Urbana-Champaign — **Expected Graduation: December 2026**
B.S. in Statistics and Computer Science, Minor in Mathematics (GPA: 3.96/4.00) — *Champaign, IL*

Technical Skills

Languages: Python, C, C++, Java, SQL, R, TypeScript, JavaScript, Bash
ML/AI: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Hugging Face, LangChain, OpenCV, Spark, MLflow
Infrastructure: Docker, Kubernetes, AWS, OCI, Terraform, Jenkins, Databricks, Git, CI/CD, MLOps
Development: REST APIs, Microservices, System Design, Linux, Agile/Scrum, React, Next.js, Flask, FastAPI
Interests: Drumming, Coin Collecting, LEGO, Traveling, Strength Training

Publications

MonitorBench: A Comprehensive Benchmark for Chain-of-Thought Monitorability in Large Language Models
H. Wang, Y. Sun, B. Ko, **Mann Talati**, et al. | *Accepted to COLM 2026* (Preprint)

Experience

Dow — **August 2026 – Present**
Data/Platform Engineer Intern (Incoming Fall 2026) — *Champaign, IL*

University of Illinois Urbana-Champaign — **January 2026 – Present**
AI Research Assistant — *Champaign, IL*
- Building dual-objective **CoT** monitorability benchmarks for **ASTRAL Lab**, extending **OpenAI**'s published methodology
- Curating datasets and **benchmark** specifications for hidden-objective detection across **19 tasks** and **7 models**
- Designing follow-up experiments to isolate prompting techniques that reduce monitorability and characterize how evasion emerges

Oracle — **May 2026 – August 2026**
Software Engineer Intern — *Santa Clara, CA*
- Built **Jupyter** notebook audit logging into **OCI**, giving government customers version history and access records
- Instrumented **20+** user-attributed audit events across **ARM** and **x86** builds, guaranteeing identical coverage
- Validated release readiness with **105+** automated tests, promoting the feature from dev through integration toward production

Ameren — **November 2024 – August 2025**
Data Science Intern — *Champaign, IL*
- Implemented load forecasting models at **under 4%** MAPE and **$50K+** savings, tuned via Ray Tune for **1%+** accuracy
- Built a Jenkins **CI/CD** pipeline with **Terraform** to promote notebooks and models across **3 environments** into production
- Added **MLOps** model and data drift monitoring with **Databricks MLflow** for the **3-model ensemble** used in the final forecast

Projects

Autonomous Driving (Repository) | *Python, PyTorch, OpenCV, Matplotlib, nuScenes* — **March 2026 – Present**
- Engineering a **perception pipeline** in **PyTorch** for **object detection**, segmentation, and **bird's-eye-view mapping**
- Building **end-to-end ML infrastructure** including data pipelines, **training workflows**, and visualization tools

AI Recycling Assistant (Repository) | *Python, AWS, TensorFlow, TypeScript* — **June 2025 – August 2025**
- Developed a recycling assistant featuring a custom CNN deployed on **AWS SageMaker** with **80%+ accuracy**
- Orchestrated a serverless **REST API** pipeline with **AWS Lambda**, **API Gateway**, and **S3** to deliver real-time recommendations

Activities

CUBE Consulting — **September 2024 – Present**
Chief Technology Officer (CTO) — *Champaign, IL*
- Automating client acquisition with targeted outreach at **35-40+** emails weekly to source new consulting projects
- Building a **Next.js** internal platform for **60+** members, unifying the point system, case resources, and project archives
- Previously led **4+ consultants** across **cross-functional** teams to design **AI-enabled AWS workflows** for client engagements

Relevant Coursework

Data Structures | Algorithms & Models of Computation | Artificial Intelligence | Deep Learning for Computer Vision | Natural Language Processing | Database Systems | Programming Languages & Compilers | Introduction to Computer Systems | Basics of Statistical Learning | Linear Algebra | Statistics and Probability II | Numerical Methods I | Statistical Modeling II